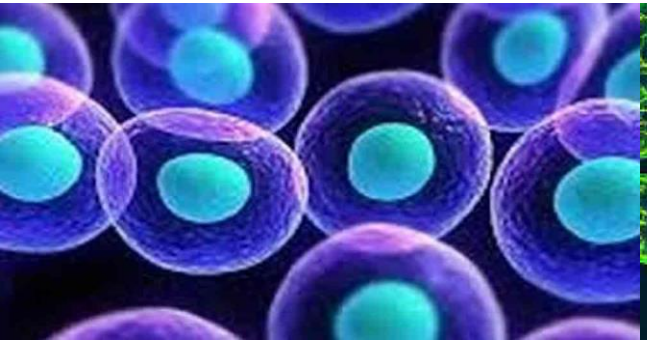
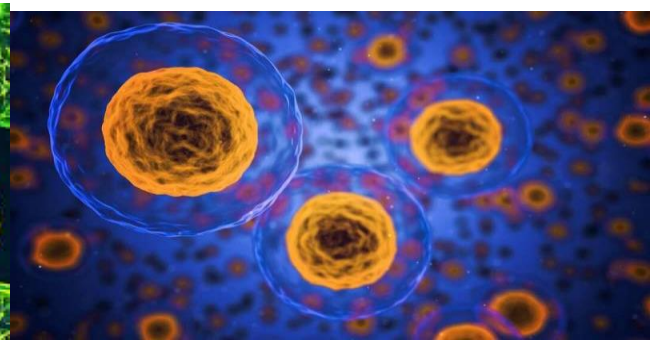
Part 8

Cell communication

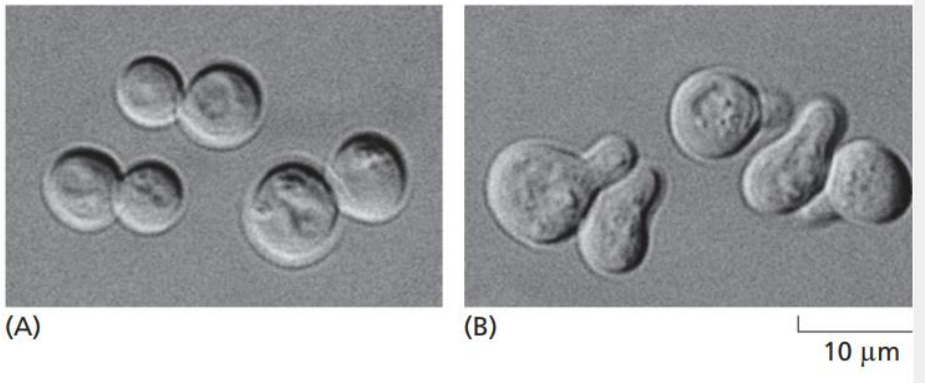




Individual cells

Need to sense and respond to their environment

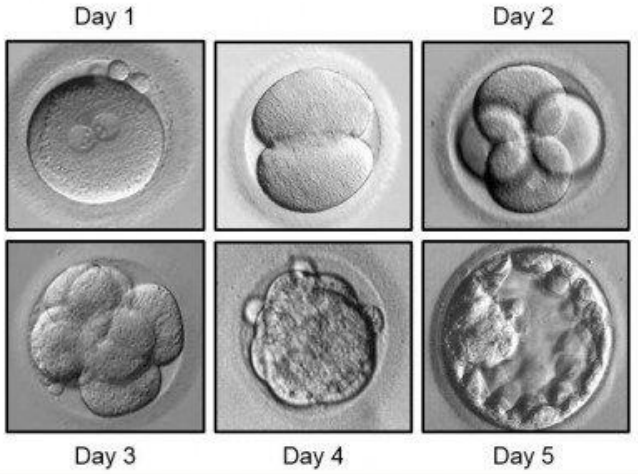
Must be able to communicate with other cells



Yeast cells respond to mating factor
Budding yeast (*Saccharomyces cerevisiae*) cells are normally spherical (A), but when they are exposed to an appropriate mating factor produced by neighboring yeast cells (B), they extend a protrusion toward the source of the factor.
(courtesy of Michael Snyder.)

Multicellular

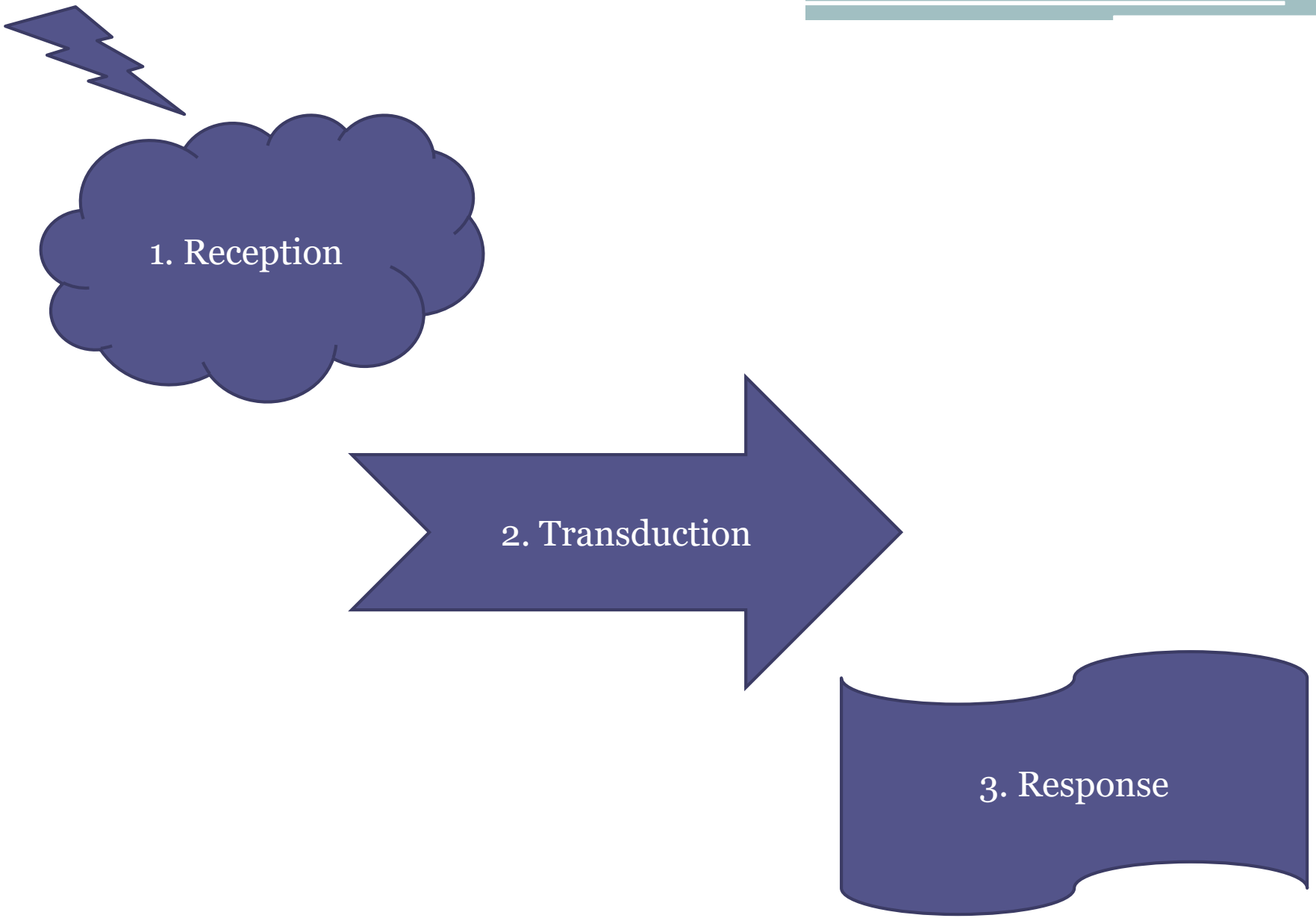
More complicated
Cells must interpret the multitude of signals → coordinate their behavior



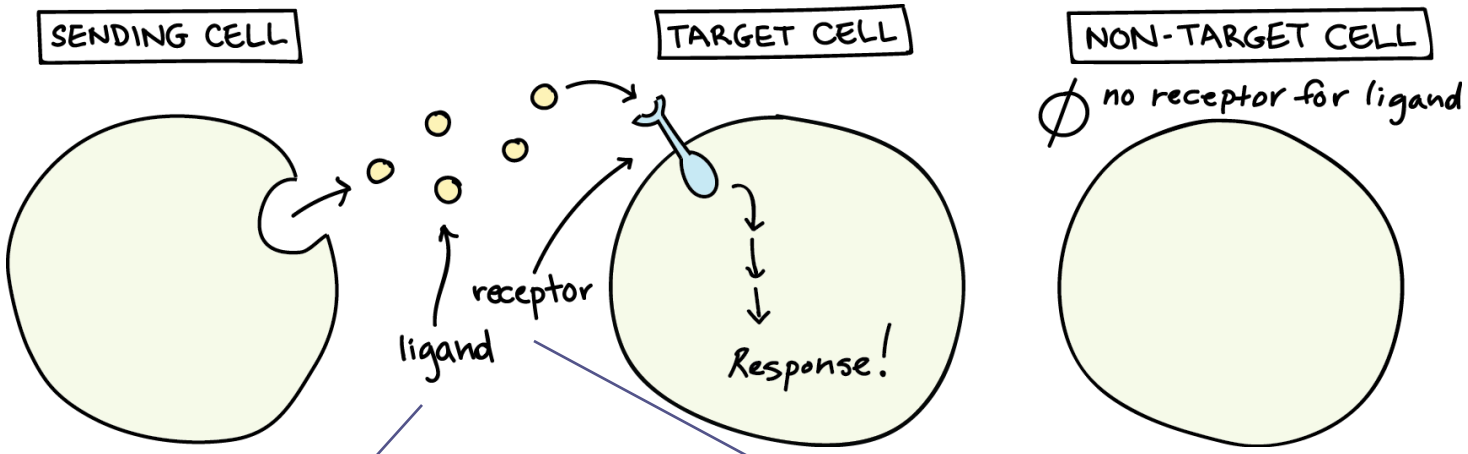
Stages of embryo development in culture

During animal development, for example, cells in the embryo exchange signals to determine which specialized role each cell will adopt, what position it will occupy in the animal, and whether it will survive, divide, or die

➔ GENERAL PRINCIPLES OF CELL SIGNALING ?


1. Reception
2. Transduction
3. Response

How can Cells detect what's going on around them, and can respond in real time?



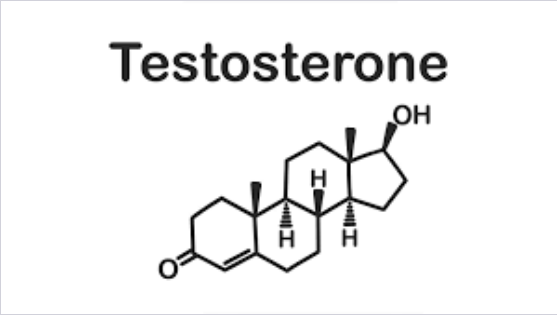
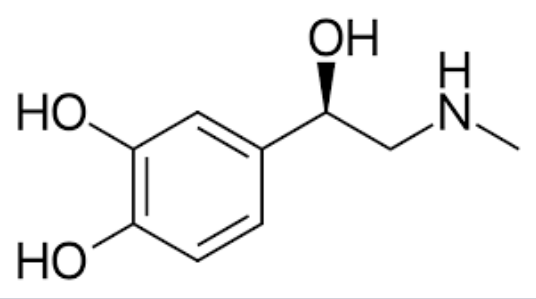
A **ligand** is a molecule that binds another specific molecule **(receptor),** in some cases, delivering a signal in the process

Receptors are protein molecules inside the target cell or on its surface that receive a chemical signal **(ligand)**

Ligand: signaling molecule

a specific ligand will have a specific receptor

Types of ligands

Lipid- based ligands	Protein-based ligands
Small, nonpolar/non soluble -> can cross the membrane	Polar/water soluble -> cannot cross the membrane
Example: Steroid hormones	Example: Protein - based, hormones derived from amino acids, polypeptides.
	Epinephrine 

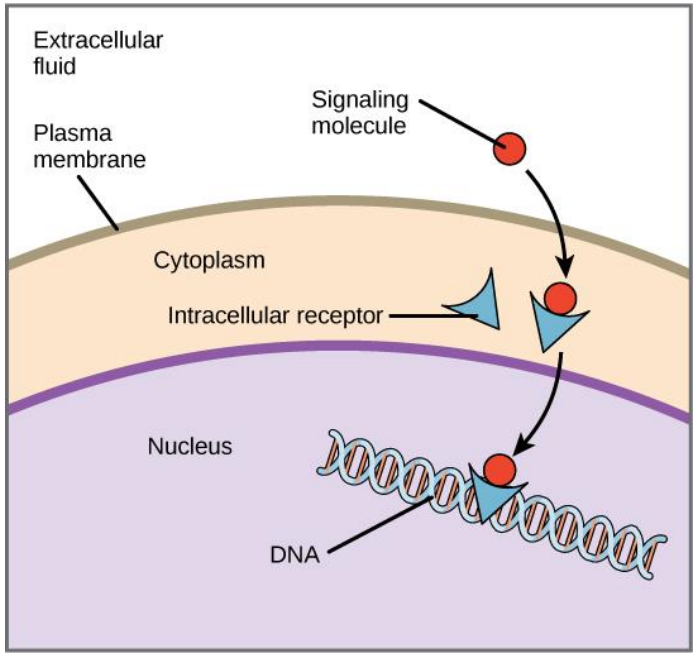
Types of receptors

Internal receptors

+ Found in the cytoplasm of the cell

+ Respond lipid-based ligands

+ Have a direct effect on protein production by binding directly to the DNA.

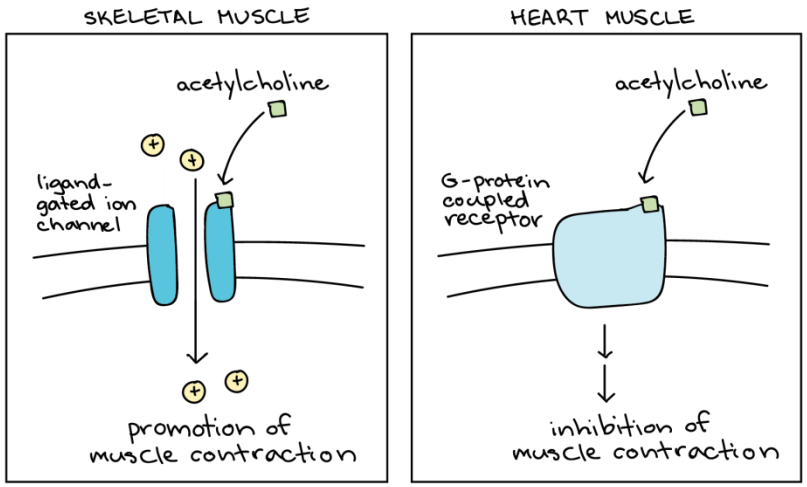


Types of receptors

Cell-surface receptors

+ Found on the cell membrane

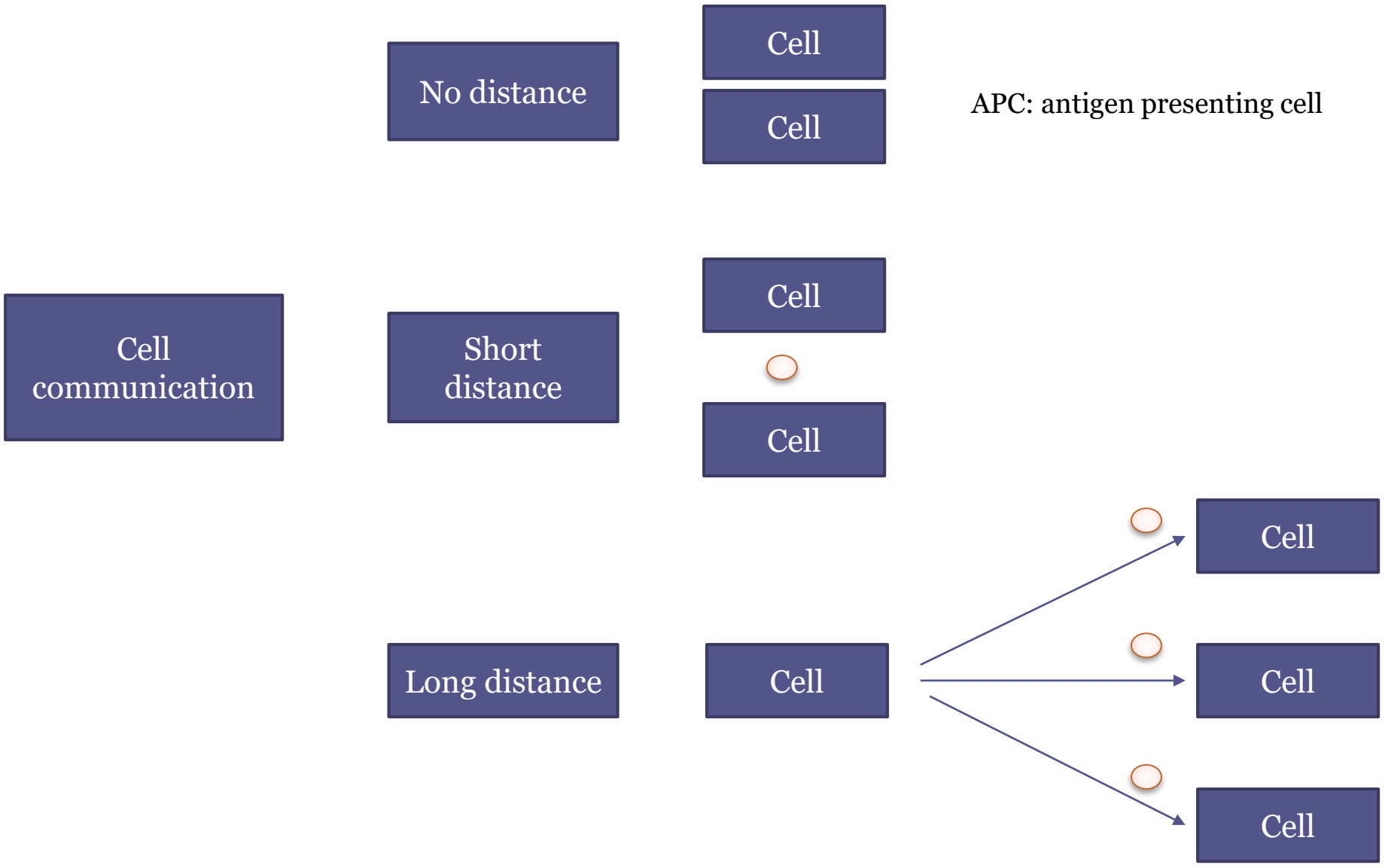
+ Bind to **protein-based ligands**



The same signaling molecule may produce different results

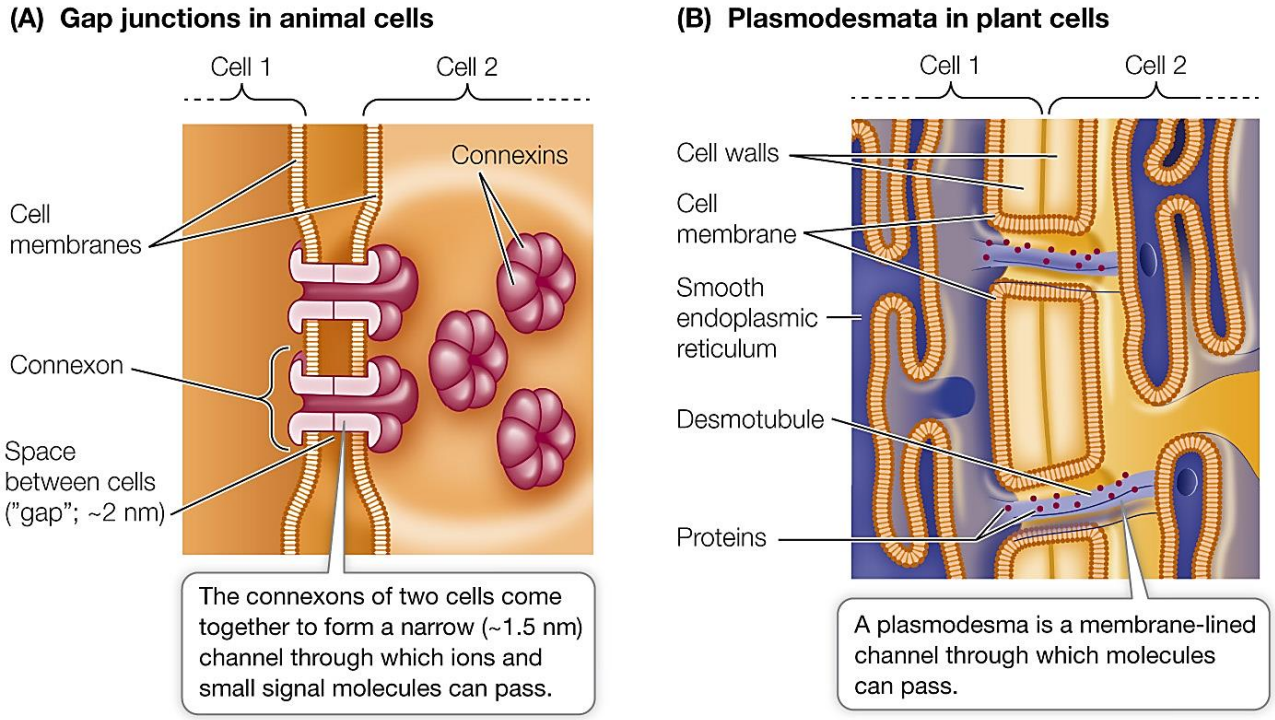
TABLE 16–1 SOME EXAMPLES OF SIGNAL MOLECULES

Signal Molecule	Site of Origin	Chemical Nature	Some Actions
Hormones			
Adrenaline (epinephrine)	adrenal gland	derivative of the amino acid tyrosine	increases blood pressure, heart rate, and metabolism
Cortisol	adrenal gland	steroid (derivative of cholesterol)	affects metabolism of proteins, carbohydrates, and lipids in most tissues
Estradiol	ovary	steroid (derivative of cholesterol)	induces and maintains secondary female sexual characteristics
Insulin	β cells of pancreas	protein	stimulates glucose uptake, protein synthesis, and lipid synthesis in various cell types
Testosterone	testis	steroid (derivative of cholesterol)	induces and maintains secondary male sexual characteristics
Thyroid hormone (thyroxine)	thyroid gland	derivative of the amino acid tyrosine	stimulates metabolism in many cell types
Local Mediators			
Epidermal growth factor (EGF)	various cells	protein	stimulates epidermal and many other cell types to proliferate
Platelet-derived growth factor (PDGF)	various cells, including blood platelets	protein	stimulates many cell types to proliferate
Nerve growth factor (NGF)	various innervated tissues	protein	promotes survival of certain classes of neurons; promotes their survival and growth of their axons
Histamine	mast cells	derivative of the amino acid histidine	causes blood vessels to dilate and become leaky, helping to cause inflammation
Nitric oxide (NO)	nerve cells; endothelial cells lining blood vessels	dissolved gas	causes smooth muscle cells to relax; regulates nerve-cell activity
Neurotransmitters			
Acetylcholine	nerve terminals	derivative of choline	excitatory neurotransmitter at many nerve–muscle synapses and in central nervous system
γ-Aminobutyric acid (GABA)	nerve terminals	derivative of the amino acid glutamic acid	inhibitory neurotransmitter in central nervous system
Contact-dependent Signal Molecules			
Delta	prospective neurons; various other developing cell types	transmembrane protein	inhibits neighboring cells from becoming specialized in same way as the signaling cell



APC: antigen presenting cell

No distance



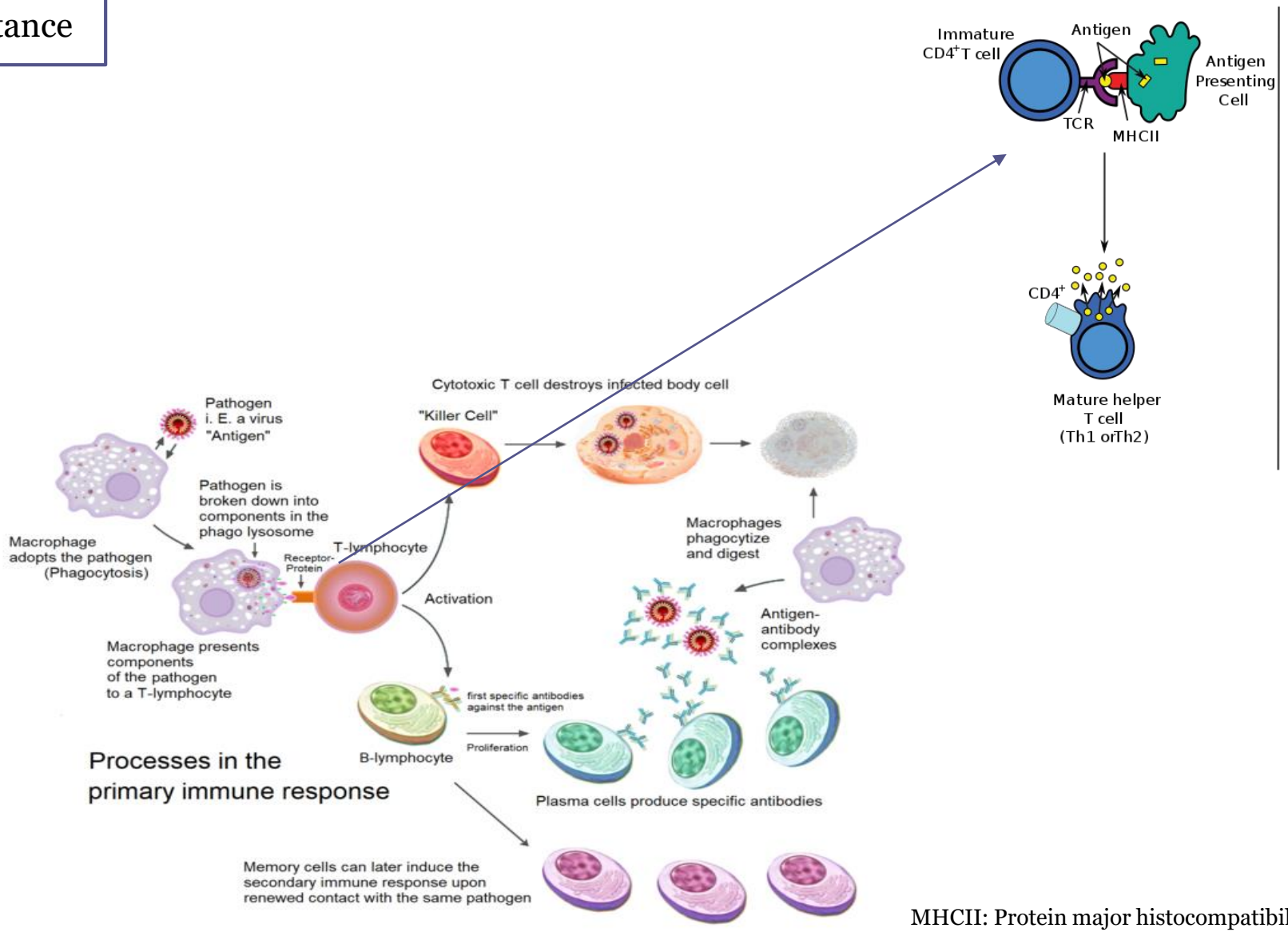
LIFE: THE SCIENCE OF BIOLOGY 11e, Figure 7.16
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Communicating Junctions

(A) An animal cell may contain hundreds of gap junctions connecting it to neighboring cells. The pores of gap junctions allow small molecules to pass from cell to cell, ensuring similar concentrations of important signaling molecules in adjacent cells so that the cells can coordinate their activities.

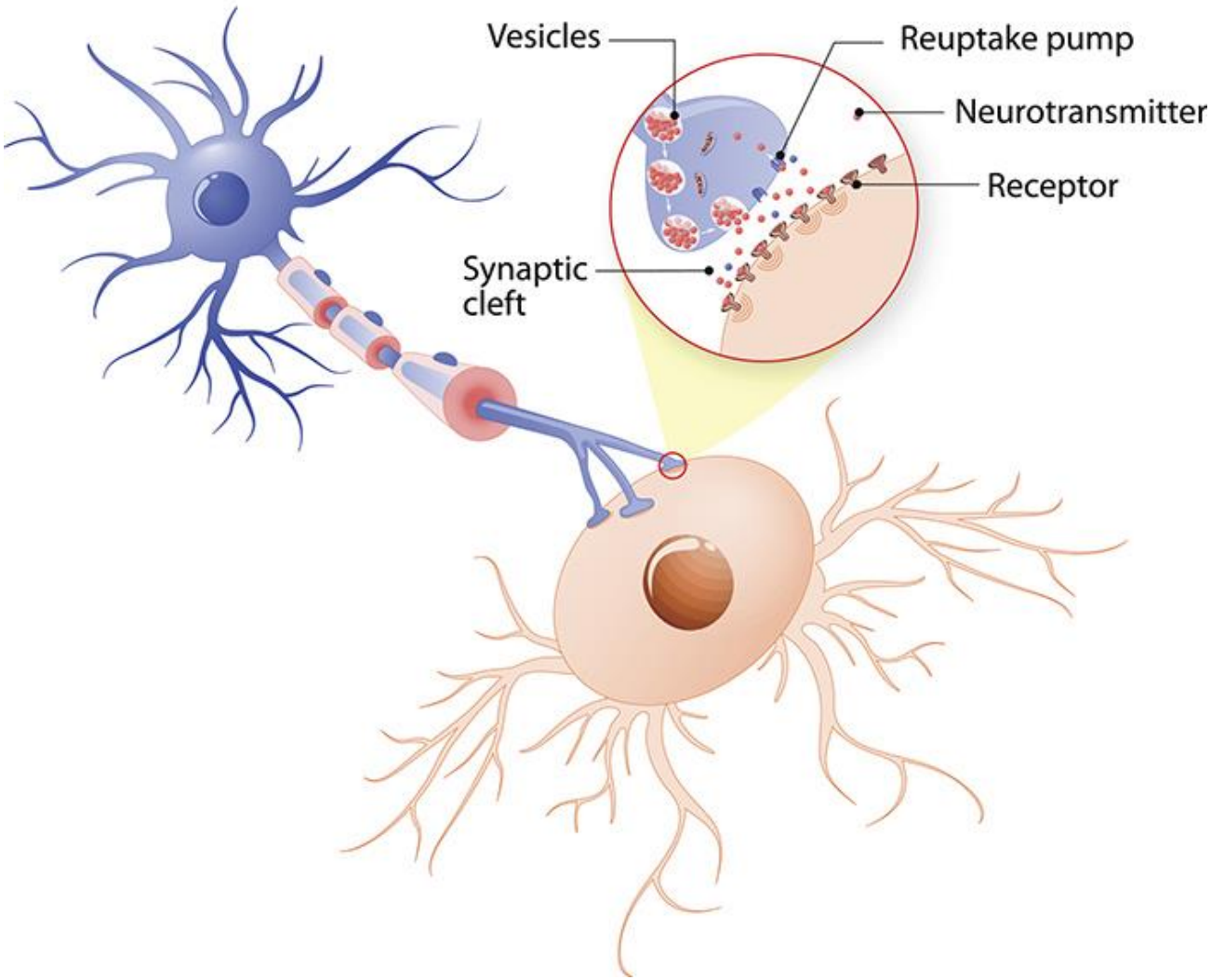
(B) Plasmodesmata connect plant cells. The desmotubule, derived from the smooth endoplasmic reticulum, fills up most of the space inside a plasmodesma, leaving a tiny gap through which small metabolites and ions can pass

No distance



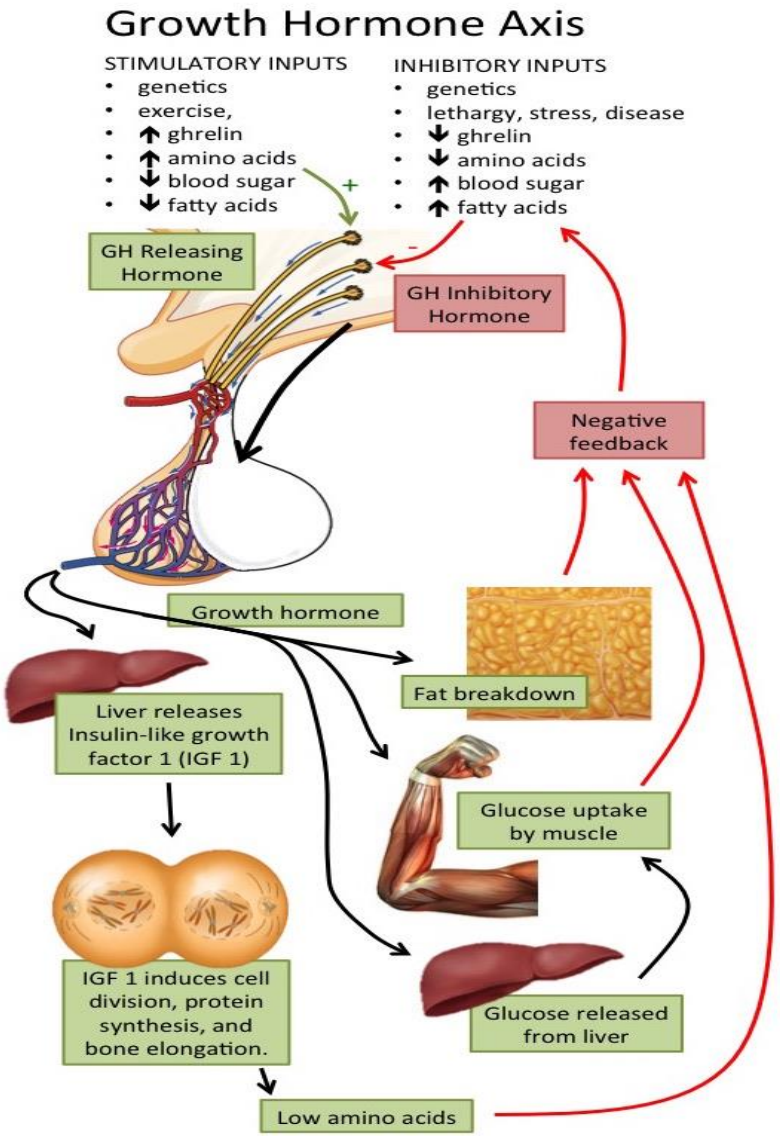
MHCII: Protein major histocompatibility complex (MHC) molecules
TCR: T cell receptors

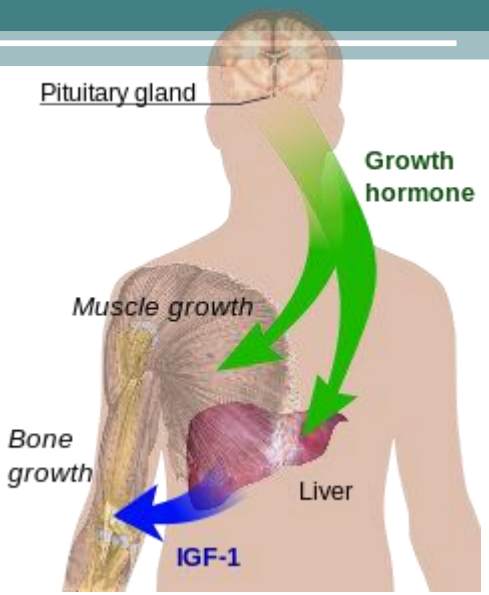
Short distance



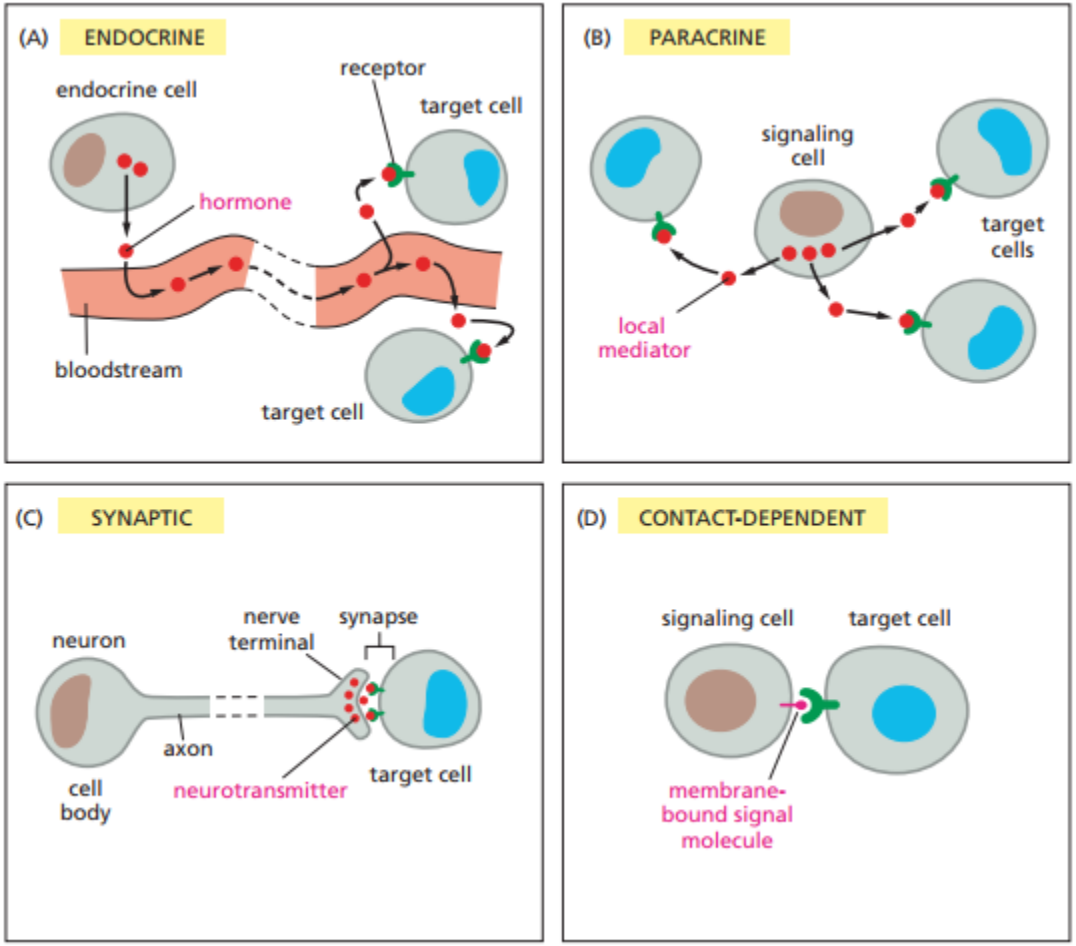
Long distance



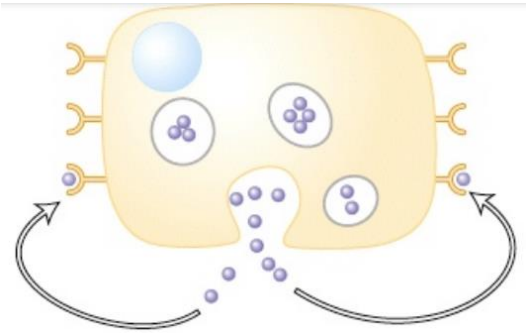




All the cell in the body receive the same message from growth hormone

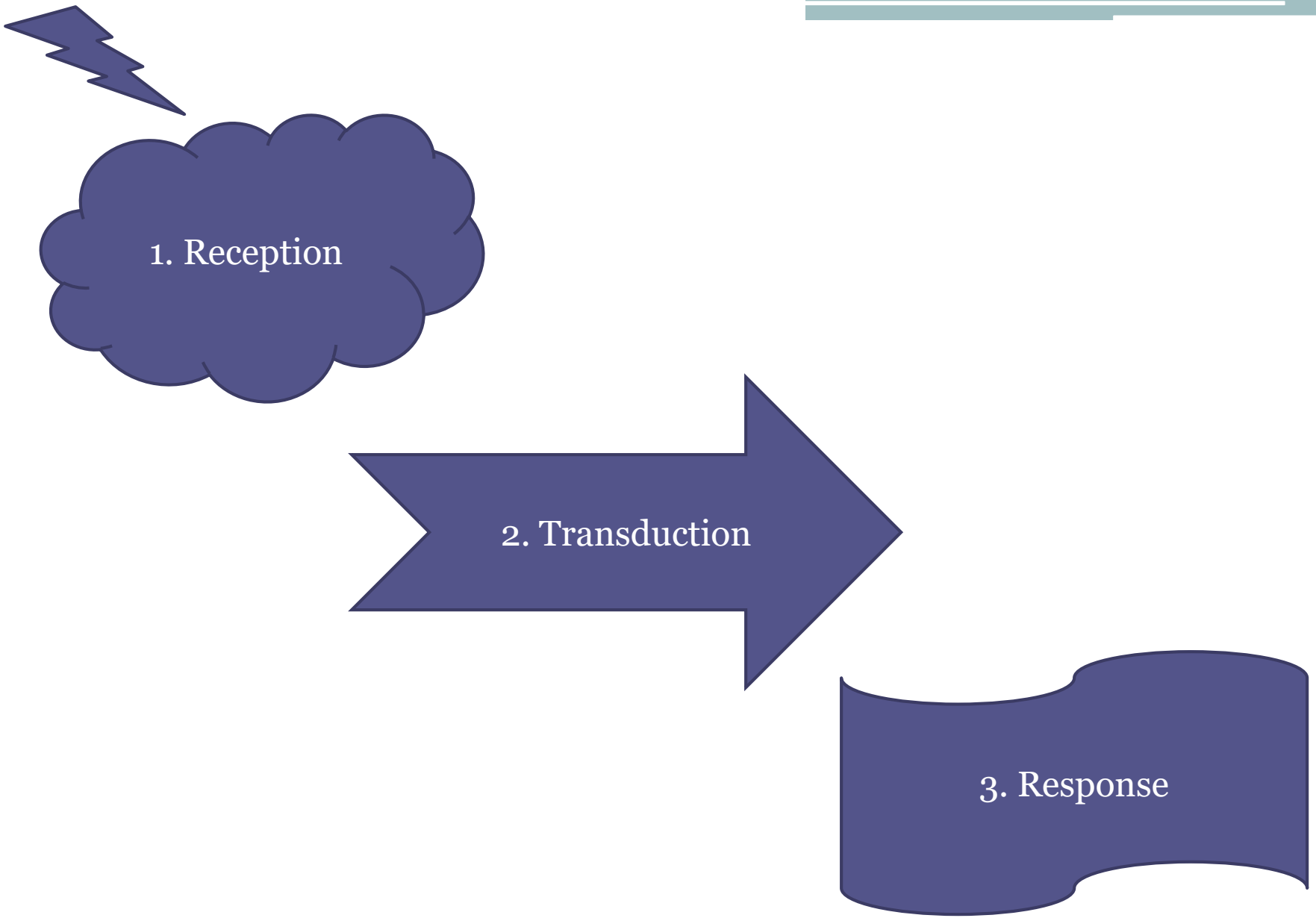


(E) Autocrine signaling



Target sites on same cell

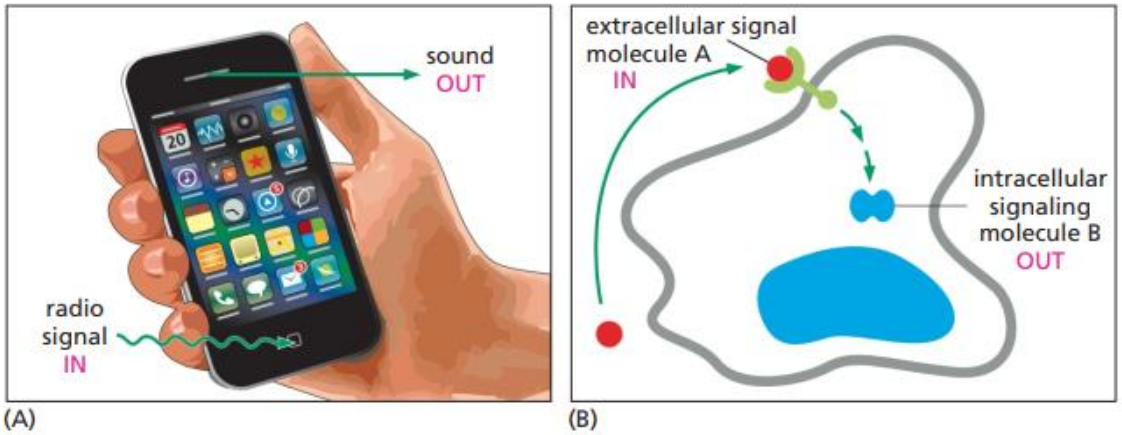
To remain a local stimulus, paracrine signal molecules must be prevented from straying too far from their points of origin. Suggest different ways by which this could be accomplished. Explain your answers.


1. Reception
2. Transduction
3. Response

Signal transduction

A signal: is a function that conveys information about a phenomenon

Signal transduction: The process whereby one type of signal is converted to another



(A) When a mobile telephone receives a radio signal, it converts it into a sound signal; when transmitting a signal, it does the reverse. (B) a target cell converts an extracellular signal molecule (molecule a) into an intracellular signaling molecule (molecule B).

Signal transduction pathway

Types

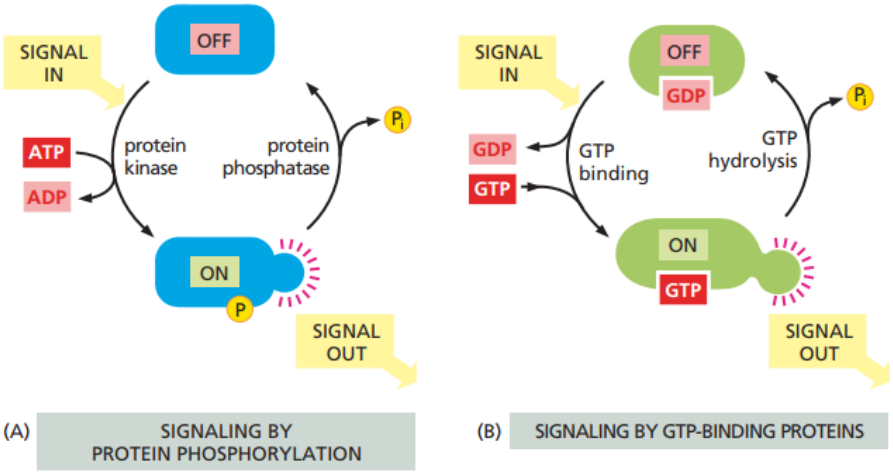
Protein modification

Phosphorylation cascade

Protein kinase

Signal transduction

*Only occurs with cell-surface receptors



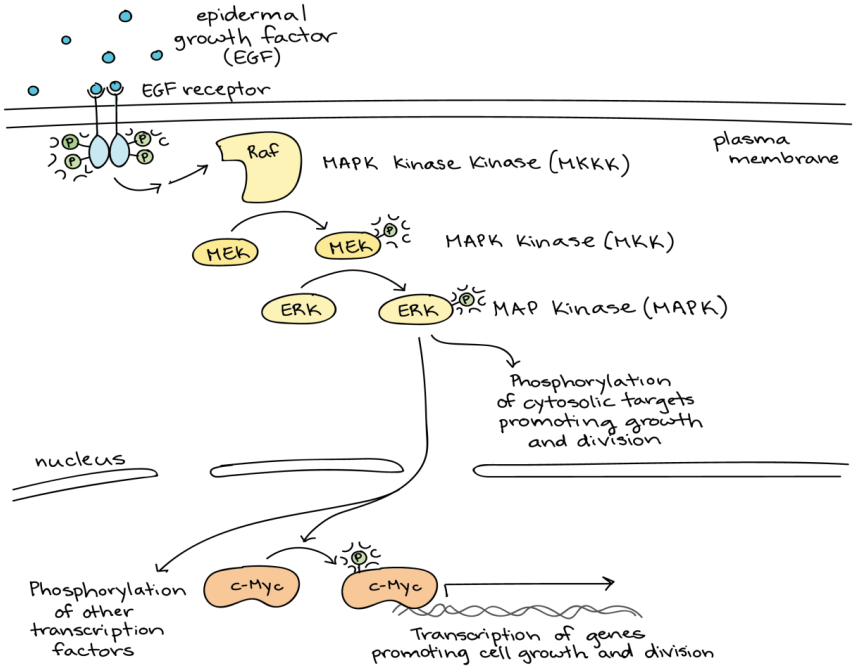
Phosphorylation

Kinases are a type of enzyme that adds a phosphate group to another molecule (including other proteins). This is called **phosphorylation.** Phosphorylation can activate or deactivate other proteins.

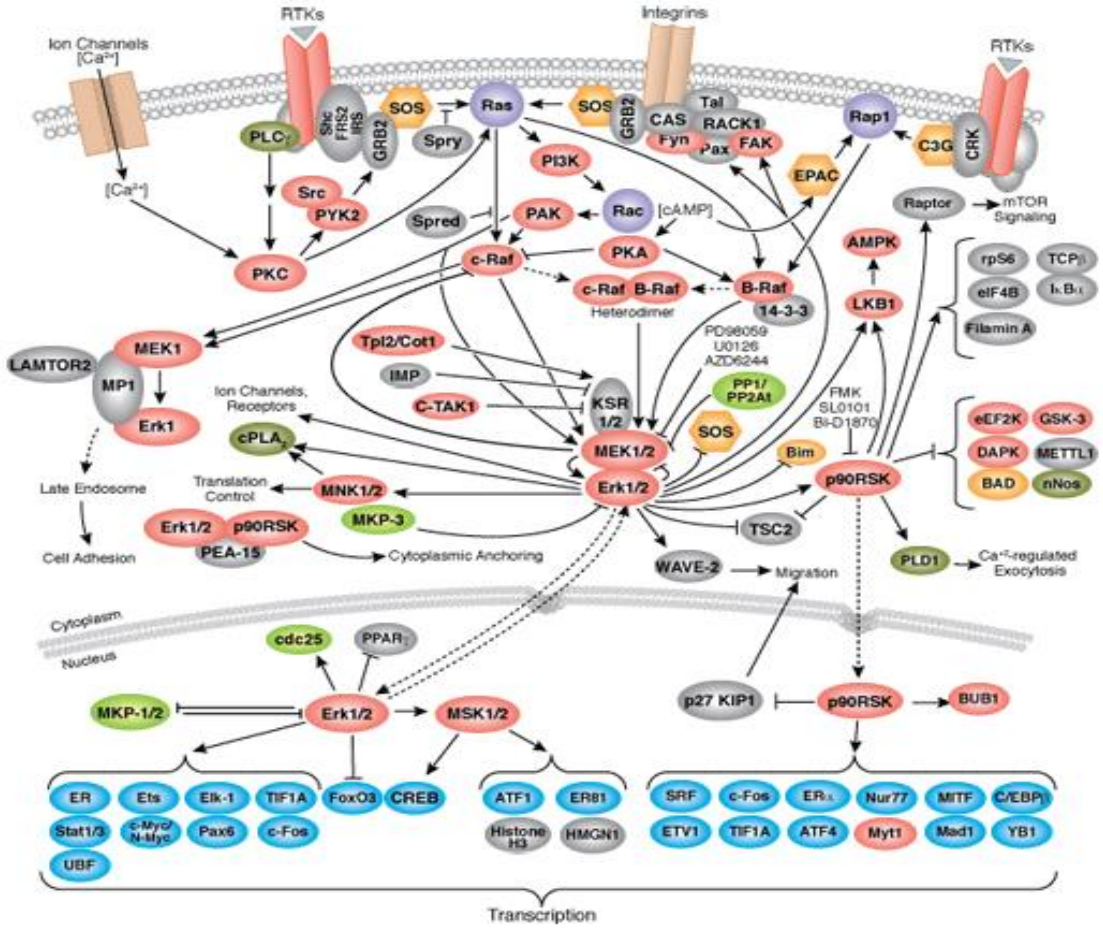
Signal transduction

Phosphorylation

Phosphorylation example: MAPK signaling cascade



MAPK/Erk in Growth and Differentiation



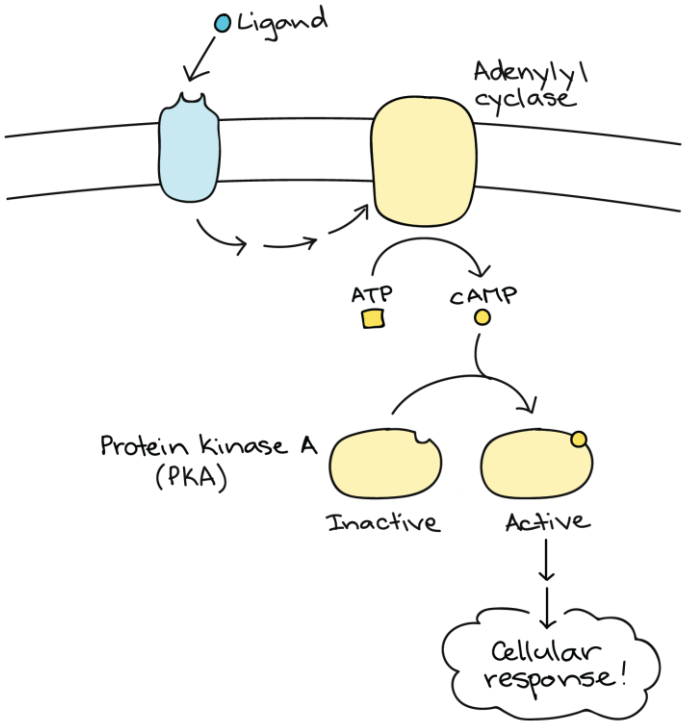
MAPK/Erk in Growth and Differentiation

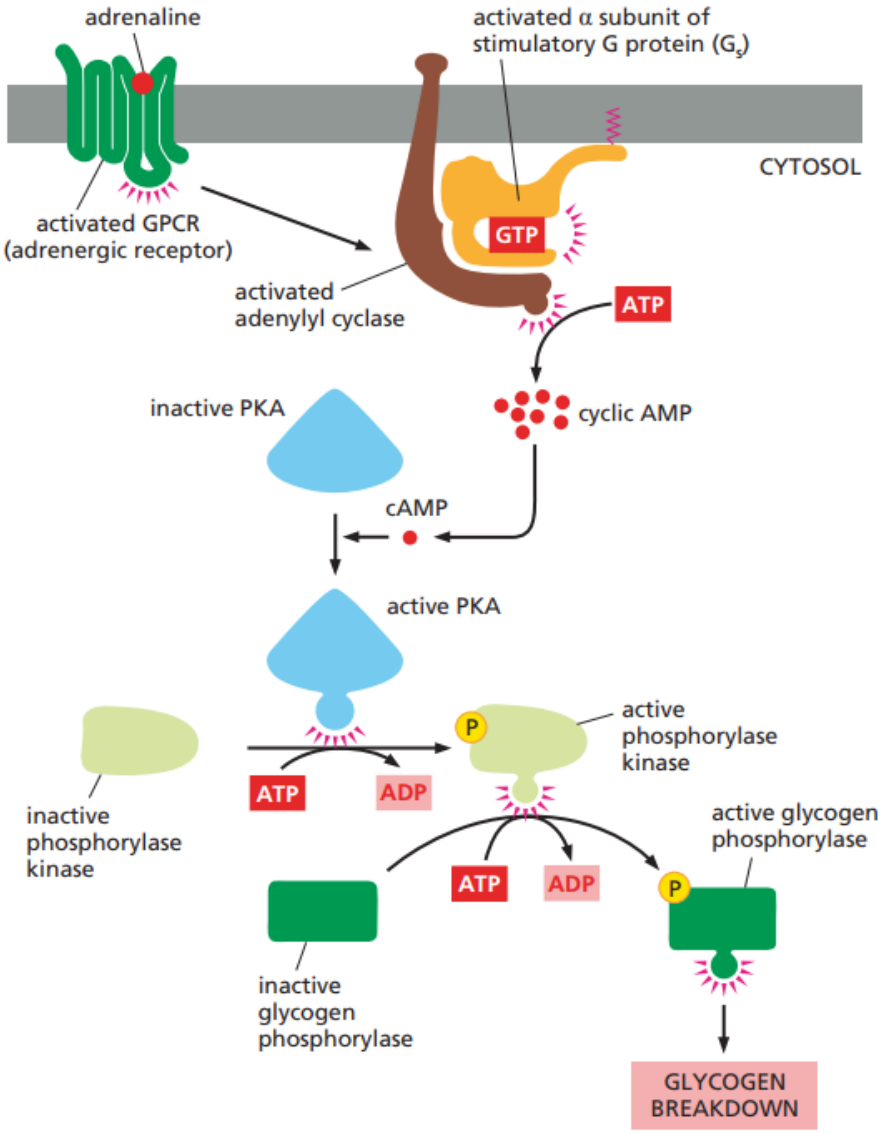
Signal transduction

Second messengers

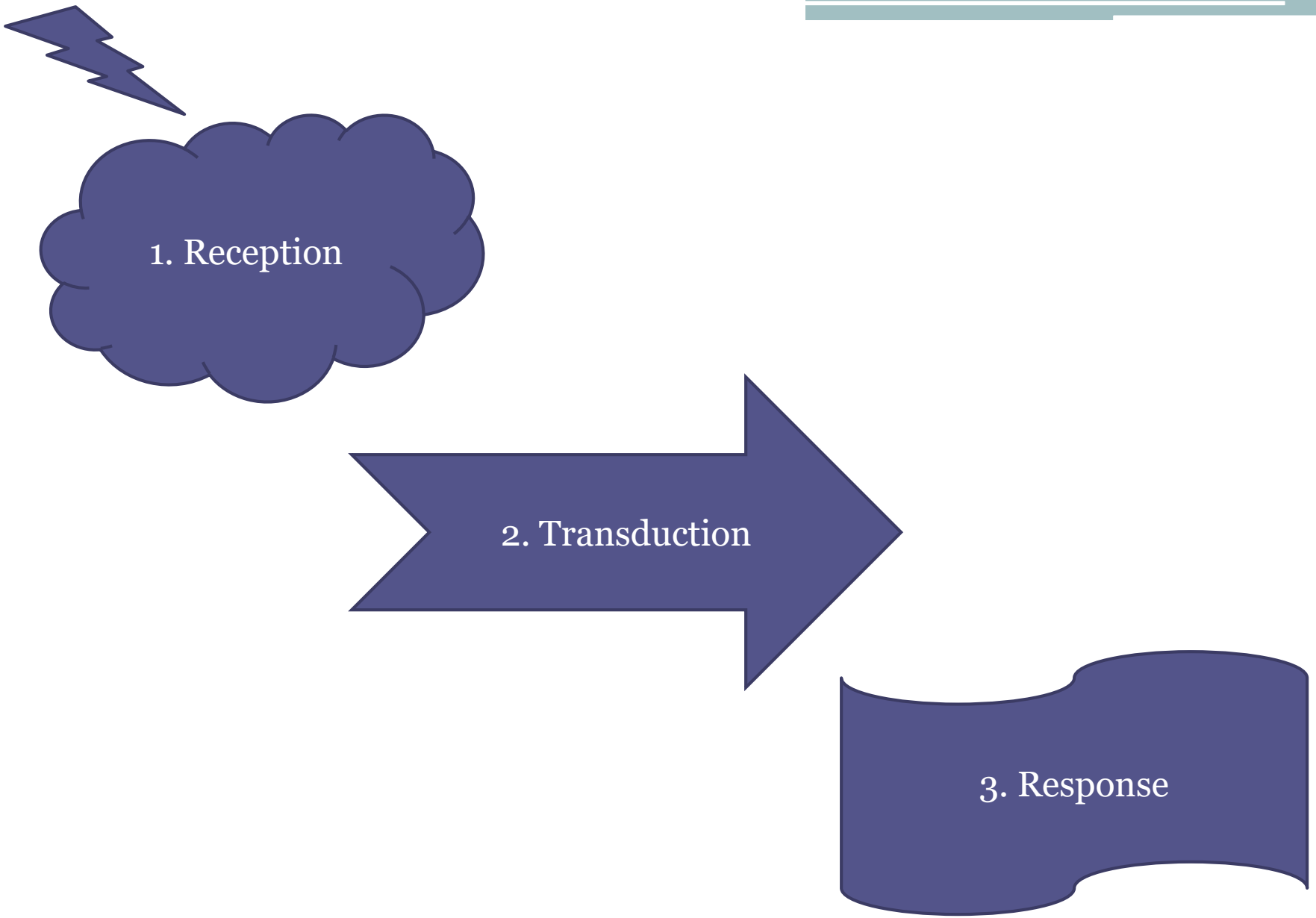
+Small molecules that help to spread a signal through the cytoplasm after a ligand binds to a receptor

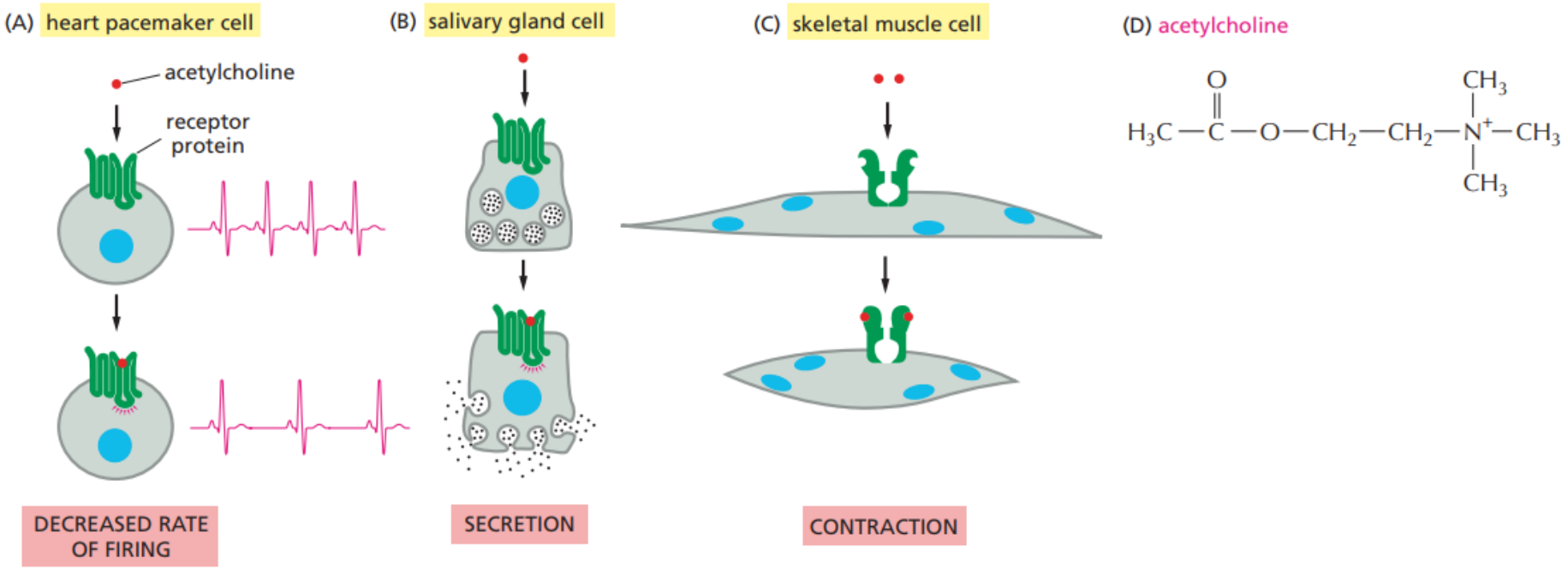
+ Ex. Cyclic AMP (cAMP); calcium ions



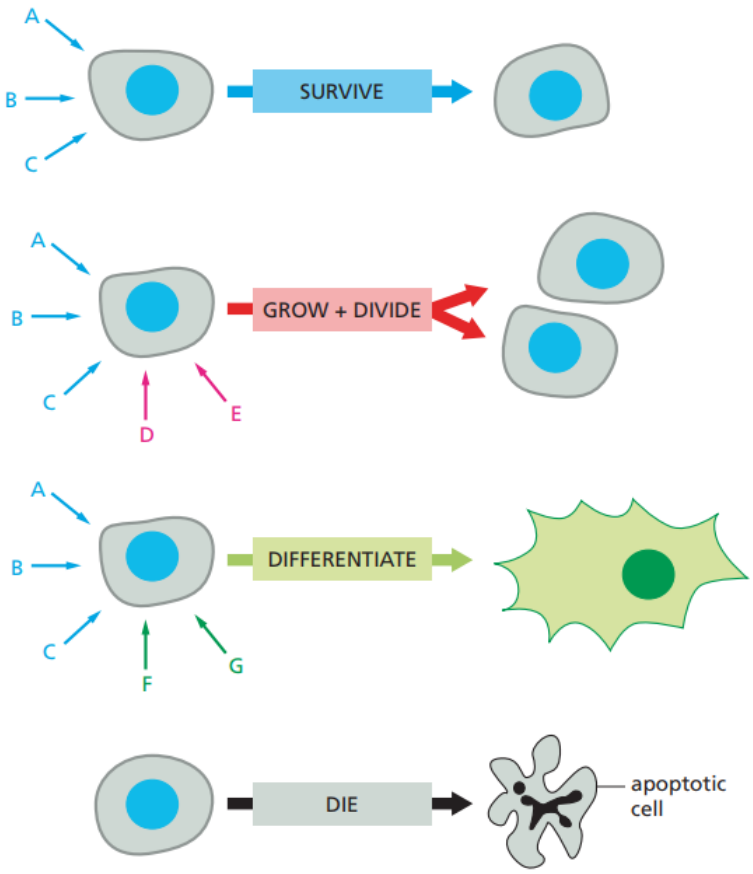


Adrenaline stimulates glycogen breakdown in skeletal muscle cells.

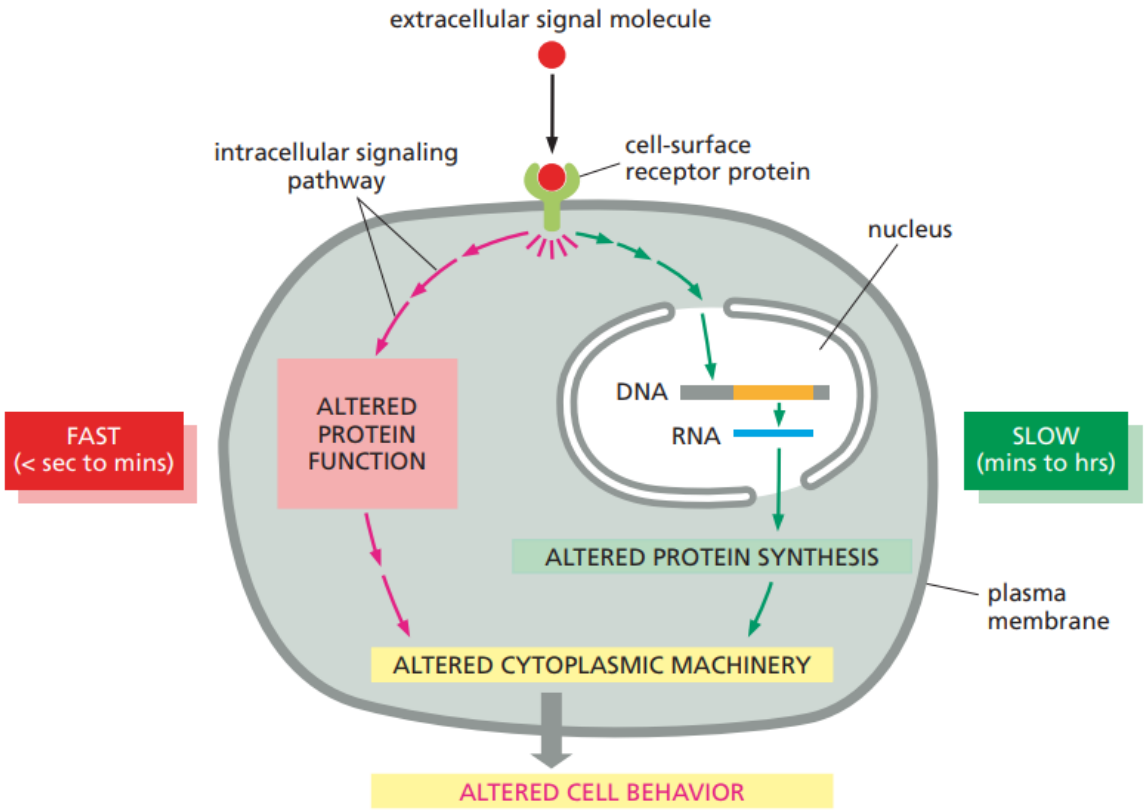

1. Reception
2. Transduction
3. Response



The same signal molecule can induce different responses in different target cells



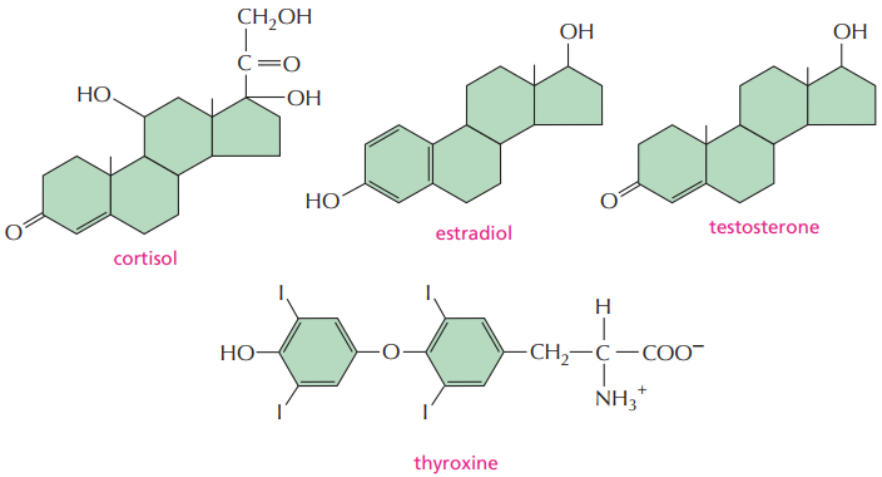
An animal cell depends on multiple extracellular signals



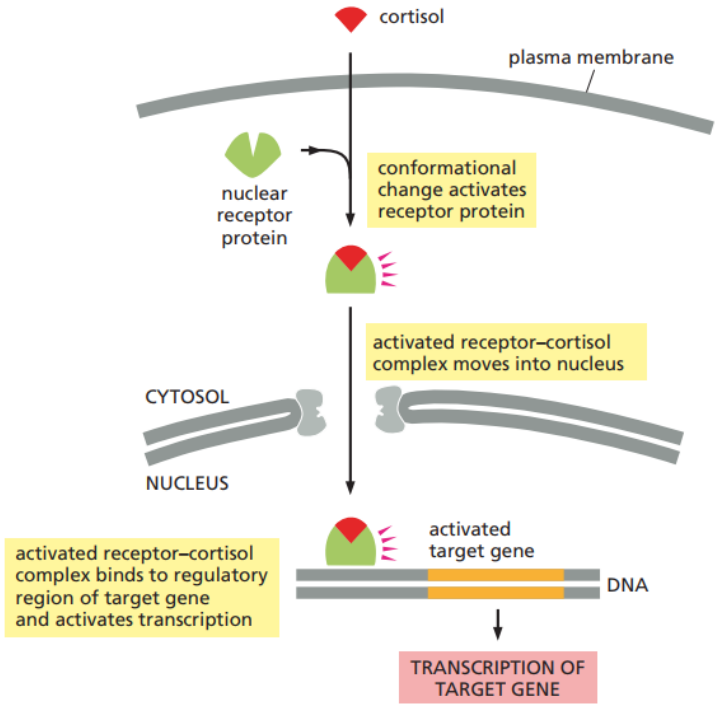
Extracellular signals can act slowly or rapidly.

Quick: changes in cell movement, secretion, or metabolism— need not involve changes in gene expression

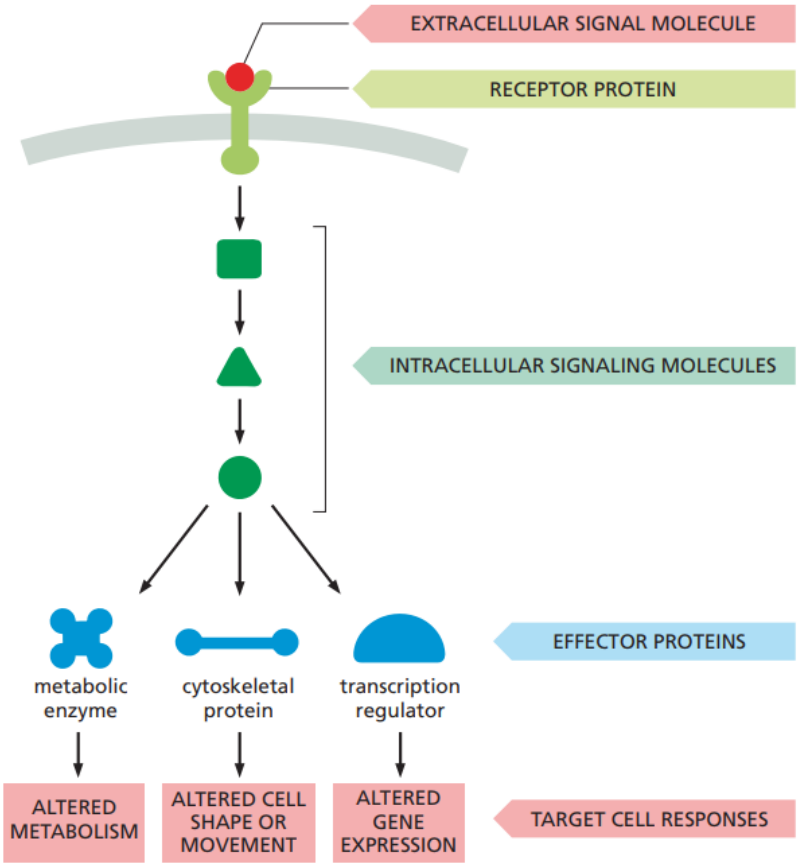
Slow: cell differentiation or increased cell growth and division—involve changes in gene expression and the synthesis of new proteins



Some small hydrophobic hormones bind to intracellular receptors that act as transcription regulators. Their receptors are not identical, but they are evolutionarily related, belonging to the nuclear receptor superfamily



The steroid hormone cortisol acts by activating a transcription regulator



Many extracellular signals act via cell-surface receptors to change the behavior of the target cell

The components of these intracellular signaling pathways perform one or more crucial functions

1. They can **simply relay** the signal onward and thereby help spread it through the cell.

2. They can **amplify** the signal received, making it stronger, so that a few extracellular signal molecules are enough to evoke a large intracellular response.

3.They can **detect** signals from more than one intracellular signaling pathway and integrate them before relaying a signal onward.

4. They can **distribute** the signal to more than one effector protein, creating branches in the information flow diagram and evoking a complex response.

Cellular responses to signals

+ **Changes in gene expression**: an increase or decrease in the production of a protein produced by a specific gene.

+ **An increase in cellular metabolism**: the conversion of glucose to glycogen (and back) can be regulated depending on the energy needs of the cell.

+ **Cell growth**: cells do not normally divide unless they are stimulated by signals from other cells.

+ **Cell death**: **apoptosis** is controlled cell death; cells can be stimulated die if they are abnormal, infected with a bacteria or virus, or during specific parts of development.

